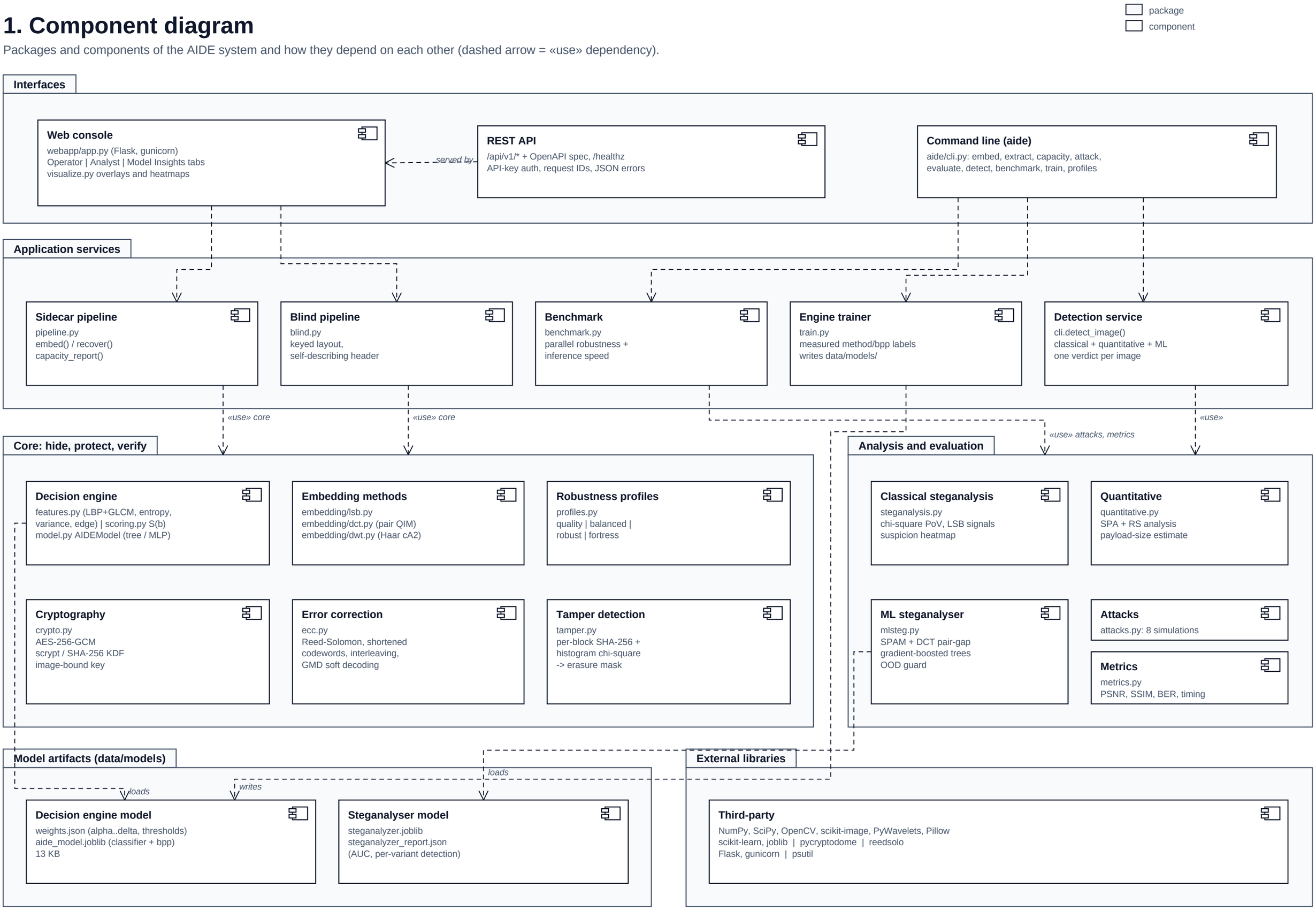
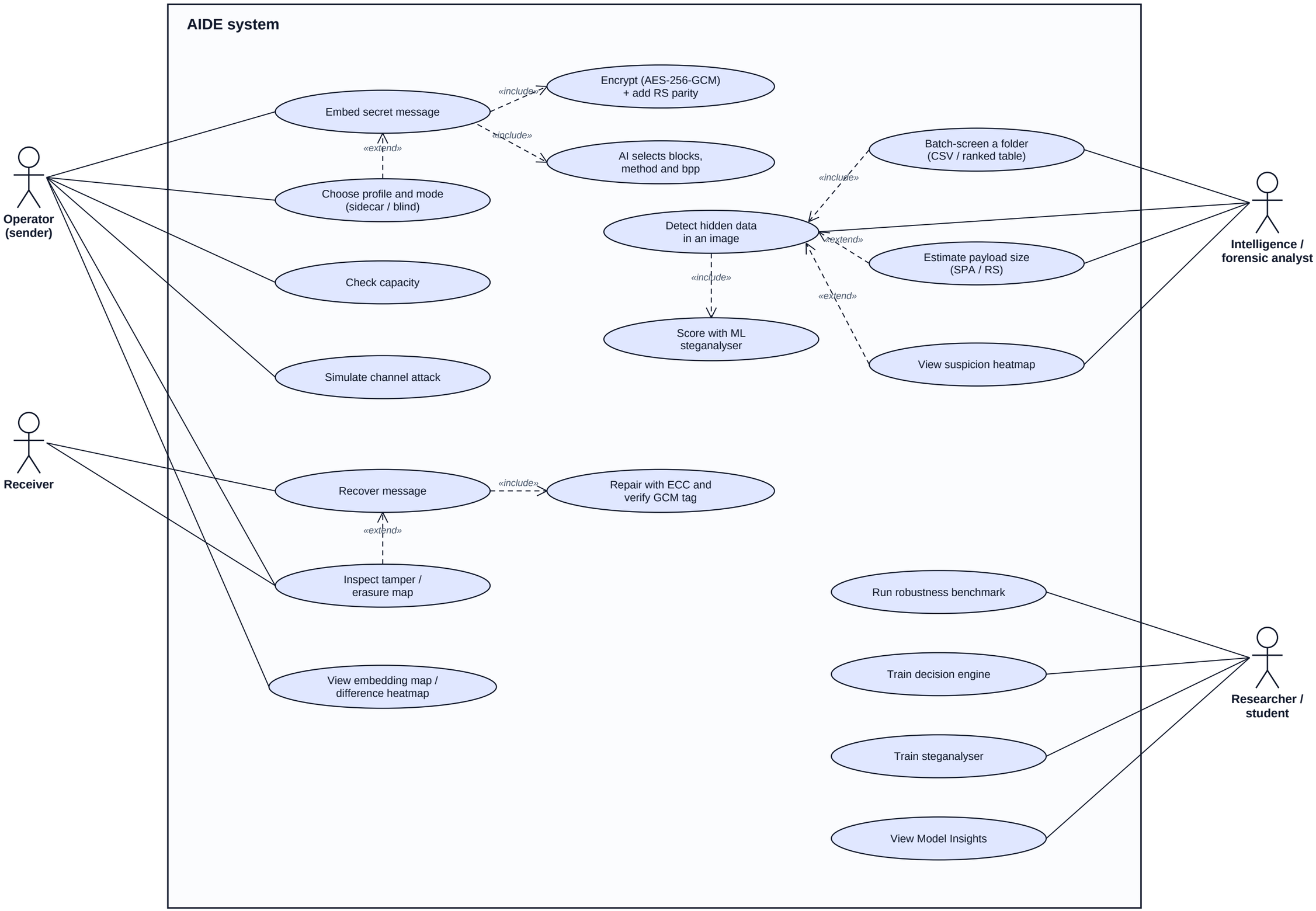
1. Component diagram

Packages and components of the AIDE system and how they depend on each other (dashed arrow = «use» dependency).



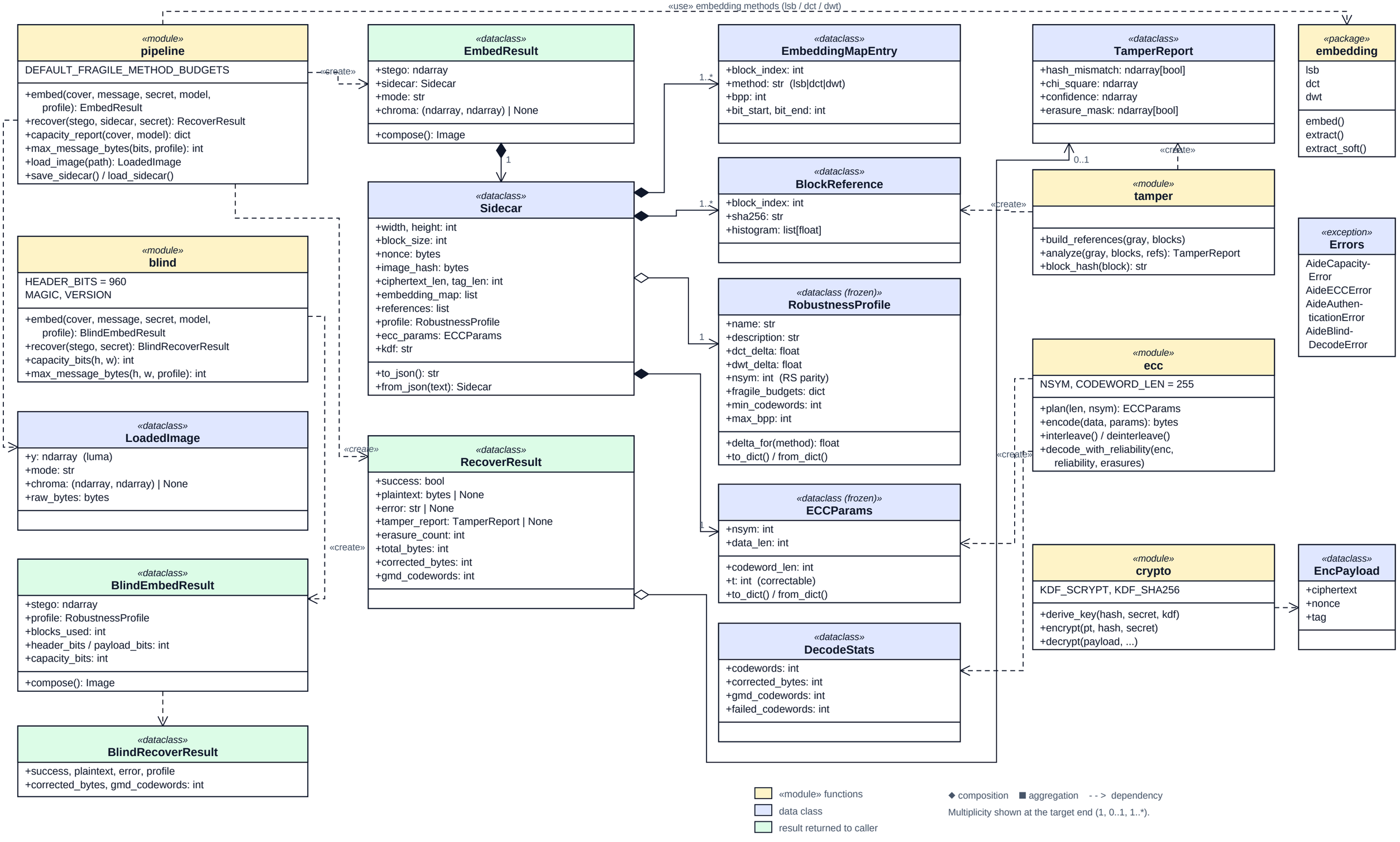
2. Use case diagram

Who interacts with AIDE and what they can do. «include» = always performed as part of; «extend» = optional extra step.



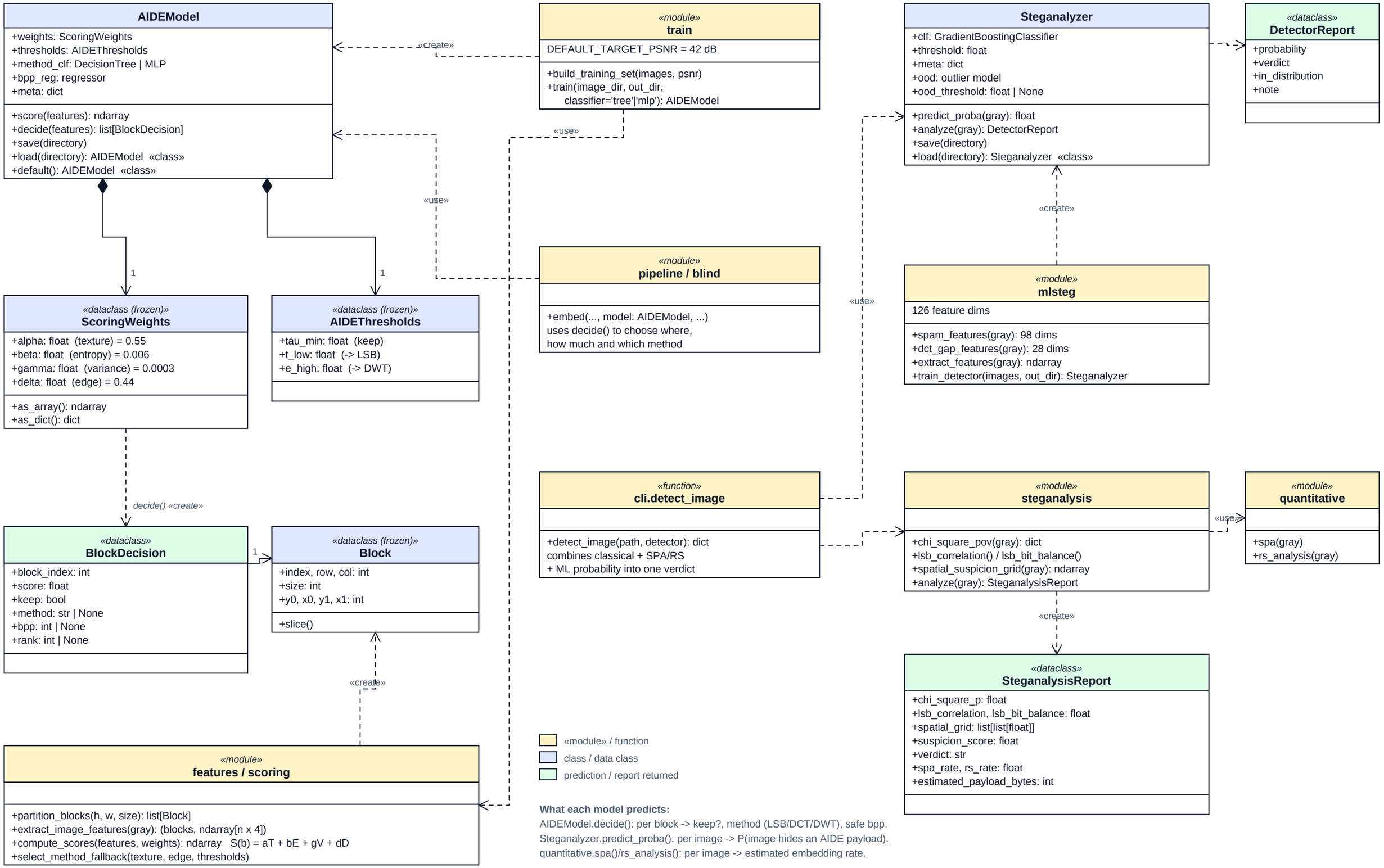
3. Class diagram: hiding and recovering a message

Main data types of the sidecar and blind pipelines (Python dataclasses). Module-level functions are shown as «module» classifiers.



4. Class diagram: decision engine and steganalysis

The trained models: AIDEModel picks blocks/method/bpp; Steganalyzer and the statistical modules analyse suspect images.

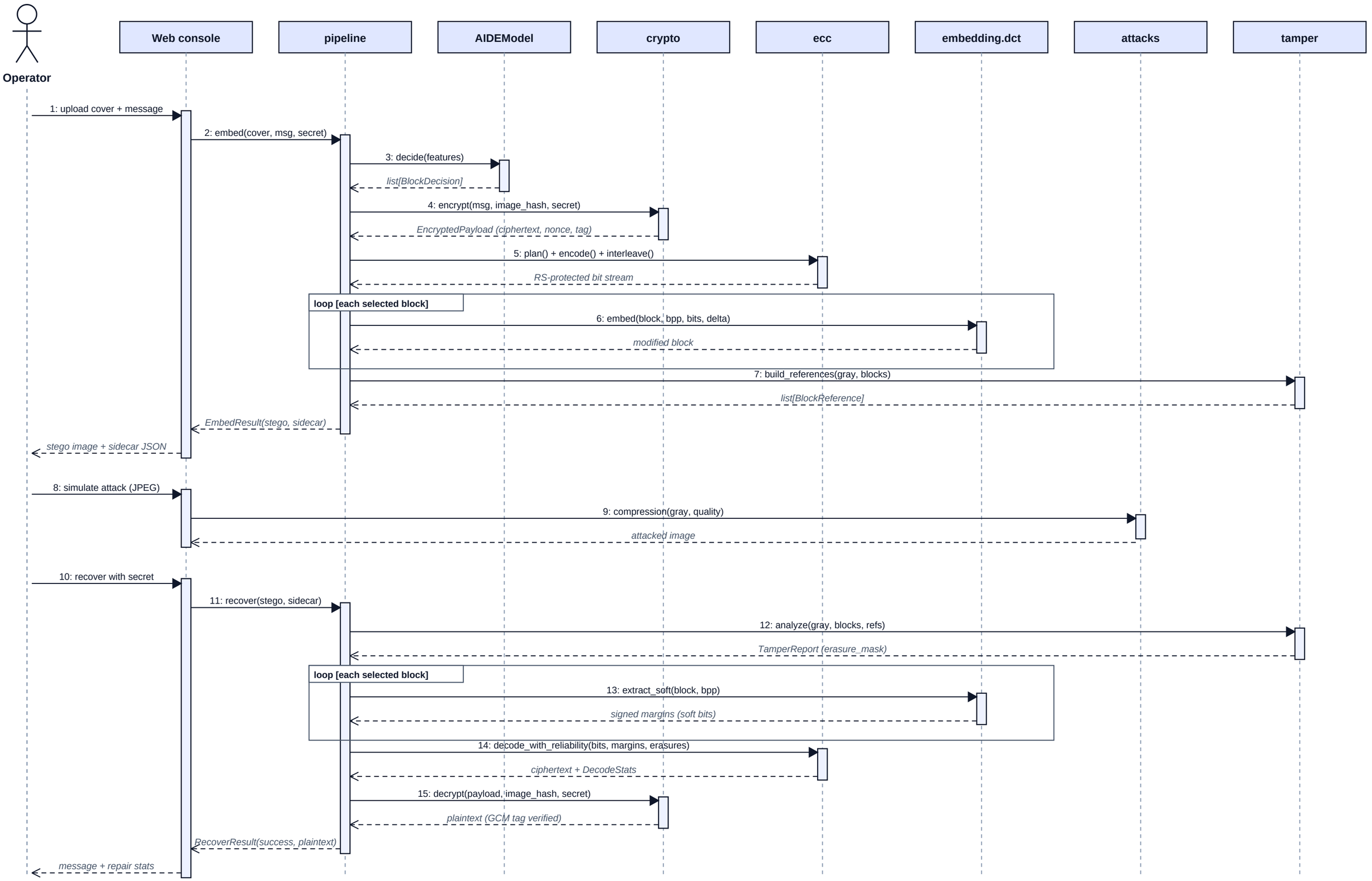


What each model predicts:

AIDEModel.decide(): per block -> keep?, method (LSB/DCT/DWT), safe bpp.
Steganalyzer.predict_proba(): per image -> P(image hides an AIDE payload).
quantitative.spa()/rs_analysis(): per image -> estimated embedding rate.

5. Sequence diagram: embed, attack and recover (sidecar mode, web console)

Solid arrow = call, dashed arrow = return. Loops run once per selected image block.



Without the right secret step 15 fails: the GCM tag check rejects it, so recovery never returns a corrupted message.